

PDblend: 选择性 P/D 分离的方法说明

Selective Disaggregation: Models, Planning, and Online Coordination

双语方法说明 / Bilingual Method Note

1 Overview: 系统概览

A shared model, a complete plan, and coordinated execution

PDblend 的目标，是在满足响应延迟要求的前提下，减少大语言模型推理的 GPU 能量开销。它不要求所有请求都分离：系统比较不同执行方式的成本，决定什么时候启用 P/D、哪些请求进入分离路径，以及给各条路径分配多少资源。本文按三个技术模块解释这一方法：离线性能与能耗建模、选择性分离规划、在线协同与分流。

PDblend aims to reduce GPU energy while meeting serving-latency targets. It does not disaggregate every request. Instead, it compares execution costs and decides when to use separate prefill and decode pools, which requests should use them, and how much capacity each path receives. Three modules provide the model, choose a complete plan, and carry it out online.

先理解 P、D 和 M。 P (prefill) 读取输入并建立 KV 缓存；在本文的接力协议中，它还产生首个输出 token。D (decode) 利用已有缓存，继续生成后续 token。M (mixed) 在同一个实例内完成这两个阶段。token 是模型处理的文本片段，不一定等于一个汉字；KV 则是后续生成可以复用的中间状态。

Prefill, denoted P, processes the prompt and creates its KV state; the carry protocol also obtains the first output token from P. Decode, denoted D, reuses this state to generate subsequent tokens. A mixed instance, M, performs both stages locally. Tokens are model text units, and KV is reusable intermediate state.

为什么需要选择性分离。 在 M 上执行可以省去跨实例交接，但较长的 prefill 可能拖慢同机的 decode。分离后，P 和 D 可以各自配置资源与频率，却要付出 KV 交接、调度以及维持不同资源池的代价。因此，长请求并非总应分离，短请求也不能仅凭长度忽略系统状态；决策要同时看负载、延迟和完整成本。

Mixed execution avoids an inter-instance handoff, but long prefills can interfere with ongoing decoding. Separate pools allow different allocations and clocks, at the cost of KV transfer, coordination, and pool capacity. Length is therefore a

Table 1. 模块接口 / Module interfaces

模块	输入 / Input	输出 / Output
建模	模型、硬件、并行配置、样本	时间、功率、开销与有效范围
规划	模型、负载、SLO、预算、当前配置	角色数、频率、停驻状态、阈值
在线	请求、Plan、负载与执行状态	token、有效配置、动作与反馈

routing signal within a workload-aware plan, rather than a universal rule that long requests must always be separated.

三个模块的职责。 离线模块回答“一个候选方案大概花多少时间、消耗多少能量”，不直接决定请求去向。规划模块把请求到达率和长度分布转为两条路径的负载，比较候选方案，输出一份完整 Plan。在线模块把 Plan 落实到就绪实例，并持续收集观测；有风险时，快速保护可以提前触发新一轮规划。

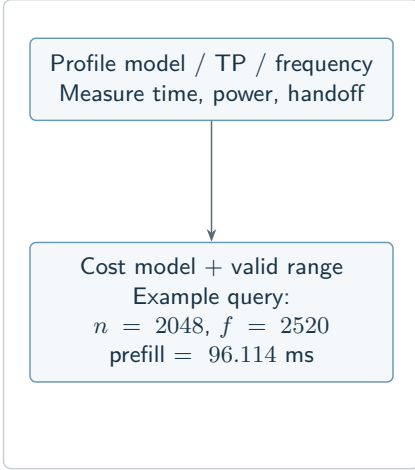
The model estimates candidate costs without routing requests. The planner translates demand and length statistics into the work assigned to each path, compares candidates, and emits a complete Plan. The runtime applies the Plan to ready instances and reports observations. Protection can request replanning before the next periodic deadline.

Plan 为什么必须完整。 一个阈值无法独立描述分离方案。阈值改变后，两边需要的容量也会改变。因此，Plan 一起指定 P/D/M 实例数、各池频率、停驻状态与长度阈值。新阈值只有在对应路径容量就绪后才能生效。已开始执行的请求保持自己的路径，不能因下一轮计划改变而被悄悄迁移。

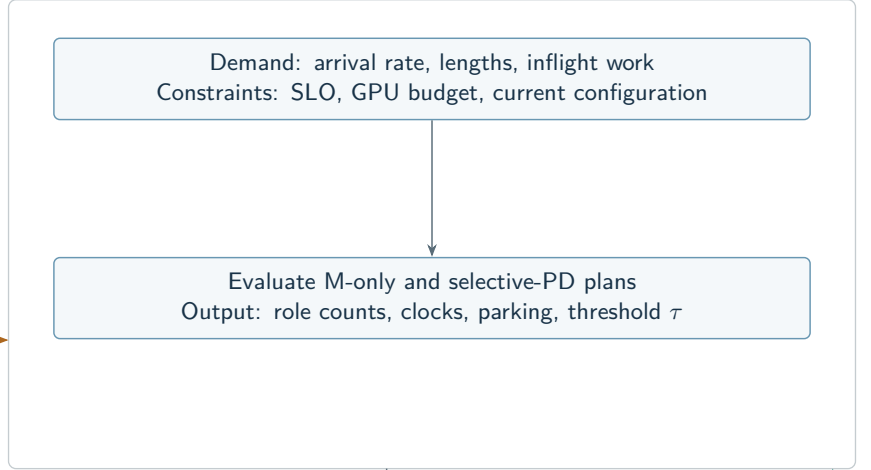
A threshold alone does not define a disaggregated configuration: moving requests changes the capacity each path needs. A Plan therefore binds role counts, clocks, parking states, and the threshold. A new threshold becomes effective with sufficient ready path capacity. Ongoing requests retain their execution ownership across plan changes.

统一的请求规格，不混淆各图假设。 下文使用 8 个 TP1 实例，每个实例占一张 GPU。短输入为 256 或 512 token，

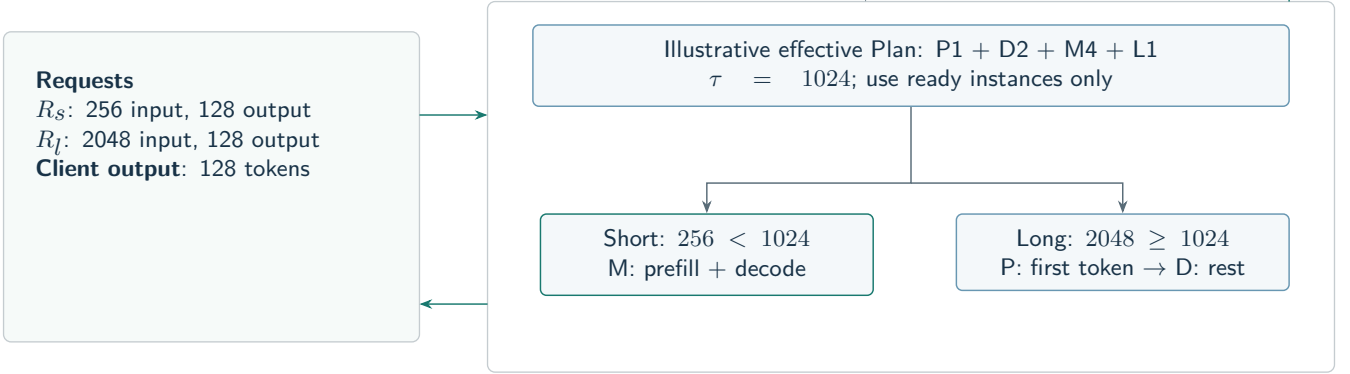
(a) Offline model supply



(b) Periodic planning



(c) Online coordination and execution



Feedback: arrivals $\rightarrow$ forecast; first token / finish $\rightarrow$ latency protection. No automatic model retraining.

Figure 1. 三个模块通过模型查询、完整 Plan 和在线观测形成闭环。图中的八实例配置与请求是协作机制示例，不是某次最优规划的测量结果。实线区分模型供给与执行流，虚线为观测反馈。

Three modules exchange model queries, complete plans, and online observations. The eight-instance layout and requests illustrate coordination, not a measured optimal plan. Dashed arrows carry feedback; the offline model is not automatically retrained by this loop.

长输入为 2048 或 4096 token，输出预算均为 128。阈值示例为 1024。图 3 演算三个候选的比较规则；图 4 另用一份已生效的 PD 配置解释执行，不把后者当作前者的最终输出。

Examples share eight TP1 instances, one GPU per instance, prompt lengths of 256, 512, 2048, or 4096 tokens, and an output budget of 128. A threshold of 1024 separates the prompt groups. Figure 3 examines selected candidates; Figure 4 uses a separate effective PD layout to explain execution. Their configuration assumptions are stated independently.

token 的平均耗时。SLO 是对这些延迟的服务目标。本文能量范围为指定 GPU 的板卡能量，包含仍有静态功率的停驻或关闭实例，不等于主机整机能量。模型预测、合成例子与硬件实测需要分别标注。

TTFT ends when the client receives the first token. TPOT describes the mean time per subsequent output token. SLOs specify the latency targets. Energy covers the selected GPU boards, including inactive devices with nonzero static draw; it is not whole-host energy. Model estimates, synthetic demonstrations, and hardware measurements remain distinct.

怎样读延迟与能量。 TTFT 是从请求提交到客户端收到首个 token 的时间；TPOT 是首 token 之后生成一个

本文的设计口径。 正文描述目标 selective PD 设计，各技术章末用短注说明当前实现差异。数值查询统一来自

仓库的合成测试模型，可由附带脚本复算；该模型直接构造系数，不能充当真实采样、拟合精度或正式评测通过的证据。详细源代码版本、计算输出和构建依赖随源码包保存。

The main text describes the target selective-PD design; implementation notes identify current gaps. Numeric queries use one reproducible synthetic fixture whose coefficients are constructed directly. It provides no evidence of hardware profiling accuracy or formal acceptance. Source identities, calculations, and build dependencies accompany the editable paper.

2 离线性能与能耗建模

Offline Performance and Energy Modeling

2.1 Motivation

同一请求可以在 M 实例上完成，也可以经 P 处理输入后交给 D 继续生成。分离能够减轻阶段间干扰，却增加 KV 交接与资源配置成本。因此，规划器需要统一的性能与能耗接口，比较完整候选方案，而不只比较某次 prefill 的速度。

A request can complete on a mixed instance or pass from a prefill instance to a decode instance. Separation can reduce interference but introduces KV handoff and configuration costs. Planning therefore needs a common performance and energy interface for comparing complete candidates, rather than isolated prefill speed.

2.2 Challenge

成本随输入长度、并发 batch、上下文长度、频率和实例布局变化。M 中的 prefill 还会影响正在解码的请求。另一方面，每条分离请求的 KV 交接与一次资源配置切换是不同事件，不能合并计费；超出模型覆盖范围的预测也不能作为可行性的证据。

Costs vary with prompt length, concurrent batch, context, frequency, and layout. Prefill on a mixed instance also affects ongoing decode work. Per-request KV handoff and resource reconfiguration are distinct events and need separate accounting. Predictions outside the model's coverage cannot establish that a candidate is feasible.

2.3 Insight

把成本拆成可复用的 prefill、decode、混合干扰、KV 交接与状态切换组成项。离线模块提供服务时间、功率及适用范围；planning 结合负载分布、实例数量和排队估计，判断哪些配置满足 SLO，以及分离是否值得。

Decompose cost into reusable prefill, decode, mixed-

interference, handoff, and state-transition components. Offline modeling provides service times, power, and validity ranges. Planning combines these with demand, replica counts, and queueing estimates to check SLOs and decide whether separation is worthwhile.

2.4 Approach

从条件到模型。固定模型、硬件及并行布局，在代表性长度、batch、上下文和频率下采样。采集稳定窗口内的 token 时序、功率和实际频率，再拟合或插值。参数冻结后，使用新采集的独立样本检查误差与覆盖范围；未通过检查的区域不能被当作可靠预测。

Fix the model, hardware, and parallel layout, then sample representative lengths, batches, contexts, and frequencies. Record token timing, power, and actual clocks during steady windows. Fit or interpolate, freeze the parameters, and check fresh independent samples. An unqualified region cannot be treated as reliable coverage.

图 2 用同一合成模型贯穿条件选择、查询与输出。该测试模型直接构造系数；图中的采样与验证链条是方法示意，表格与曲线则是可复算的模型响应，不是 GPU 实测或拟合验收结果。

Figure 2 uses one synthetic model throughout. Its coefficients are constructed directly. Sampling and validation depict the intended method; tables and curves are reproducible model responses, not GPU measurements or evidence of a completed fitting audit.

从模型到查询。实例请求包含 2048 个输入 token、128 个输出 token。假设候选稳定 batch 为 16，以 $2048 + 128/2 = 2112$ 作为代表性上下文。这两个量是候选负载假设，不能误称为实际观测。2520 MHz 下，模型返回 decode step 为 16.265 ms、功率为 127 W，因此

$$\begin{aligned} E_{\text{token}} &= \frac{\text{step time} \times \text{power}}{\text{batch}} \\ &= \frac{0.01626496 \times 127}{16} \approx 0.1291 \text{ J/token.} \end{aligned} \quad (1)$$

The example has 2048 input and 128 output tokens. Assume a candidate steady batch of 16 and a representative context of 2112. These are workload assumptions, not observations. At 2520 MHz, the model returns a 16.265 ms decode step and 127 W, giving approximately 0.1291 J per generated token across the batch.

保持接口边界。decode step 不是请求的完整延迟。KV 交接查询返回 28.488 ms，也不是用户可见 TTFT；后者截止于客户端收到 P 产生的首 token。规划器仍须计算排队、续接间隙与完整配置的能量，并计入同一 GPU 范围内的运行和停驻功耗。交接及切换只加入基础模型尚未计入的增量。

WORKED EXAMPLE: synthetic coefficients; not GPU measurements



Figure 2. 离线模型的完整查询实例。(a) 代表性采样条件及合成响应；(b) 目标建模流程与两频率响应曲线；(c) 同一请求形状的数值查询进入规划器。全部数值来自共享六频率合成模型，功率按单个 TP1 实例计；独立硬件验证并未在此图中执行。

A worked offline-model query: (a) illustrative sampling conditions and synthetic responses; (b) the modeling workflow and two frequency-response curves; (c) numerical queries passed to planning. All values use the shared six-frequency synthetic model and per-TP1-instance power. This figure does not perform independent hardware validation.

A decode step is not end-to-end request latency. The 28.488 ms handoff estimate is not user-visible TTFT, which ends when the client receives P's first token. Planning must still evaluate queueing, continuation gaps, and configuration energy, including operating and standby power over the same GPU scope. Add only overhead not already included in the base model.

实现对应 / Implementation note. 当前实现已有基础性能与功率模型、覆盖检查、KV 交接时间及部分切换成本，M 功率采用时间占比近似。统一、独立验证的增量交接能耗仍属于目标设计；本图的合成模型没有为这些

扩展提供实测资格。

Current code includes performance and power models, coverage checks, handoff timing, and partial switching costs; mixed power uses a time-share approximation. Uniformly validated incremental handoff energy remains a design target. The synthetic example does not qualify these extensions with hardware evidence.

3 成本感知的选择性分离规划

Cost-Aware Planning for Selective Disaggregation

3.1 Motivation

不同请求从分离执行中得到的收益并不相同。较长的输入可能值得交给专门的 P 池，较短的输入则可能更适合留在 M 池。规划器需要同时选择分流阈值、角色数量和频率：既让适合分离的请求获益，又保留完整部署的能量收益。这里的选择发生在两个层次：周期性比较资源方案，再按选定规则逐请求分流。

Requests benefit differently from disaggregation. Longer prompts may justify a dedicated P pool, while shorter prompts may be better served by M instances. The planner jointly selects the routing threshold, role counts, and clocks to turn these differences into deployment-level energy savings. It compares resource plans periodically; individual requests then follow the selected routing rule.

3.2 Challenge

阈值改变两条路径的请求比例，资源数量和频率又改变各自的排队与服务能力。功率更低的方案可能缺少解码容量；性能可行的方案也可能无法收回交接和配置切换的开销。因此，比较必须使用相同负载、相同能量窗口和相同 SLO 约束，还要把停驻 GPU 计入，避免只挑出某一阶段的优势。

The threshold changes the load on each path, while replica counts and clocks change queueing and service capacity. A low-power layout can lack decode capacity, and a feasible layout can still save too little to cover handoff and reconfiguration. Compare candidates under the same workload, energy horizon, and SLO constraints, including the GPUs that remain parked.

3.3 Insight

先检查候选是否有模型支持、是否具备足够容量，再比较完整窗口成本。选择性分离不是固定偏好：收益足够时采用，收益不足时保留当前可行方案。规划器把资源配置与阈值作为同一个 Plan 输出；在线层执行它，并将观测反馈给下一轮规划，而不另行改变已经评估的分流比例。

First check model coverage and capacity, then compare complete window costs. Selective disaggregation is a conditional choice: use it when it saves enough, and retain a feasible current plan when it does not. Resource allocation and threshold belong to one Plan. The runtime executes that Plan and returns observations for the next decision without independently changing the evaluated traffic split.

3.4 Approach

输入与输出 / Inputs and outputs. 输入为负载估计 F 、离线模型、SLO、GPU 预算、当前就绪配置 c 和窗口 H 。负载估计包含到达率以及输入、输出长度统计。输出 Plan 包含 P/D/M 数量、各池频率、停驻状态与阈值 τ ；执行器负责将这些决策落实为可用配置。

Inputs are demand estimate F , offline models, SLOs, the GPU budget, ready configuration c , and horizon H . Demand includes arrival rate and input/output length statistics. A Plan specifies P/D/M counts, pool clocks, parking states, and threshold τ . The executor turns that Plan into an available configuration.

比较步骤 / Candidate comparison. 枚举全 M 与 P/D/M 候选，按 τ 计算两条路径的负载。没有匹配模型支持的候选先排除，再检查时延、吞吐和 KV 容量。目标模型分别约束用户可见的首 token 延迟与 P-D 续接表现。对可行方案计算：

$$J(x) = H P(x) + E_{\text{extra}}(x, H) + E_{\text{switch}}(c, x). \quad (2)$$

$P(x)$ 包含运行与停驻功率；后两项仅计尚未包含的交接增量与本次切换增量，不重复计算基础能量或历史成本。当前方案同样重新评估，且切换到自身的成本为零。

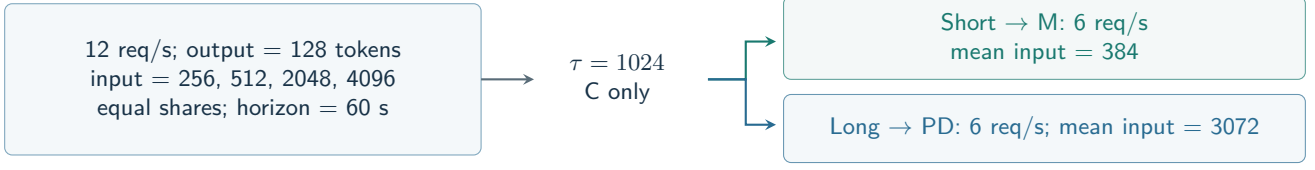
Enumerate mixed-only and P/D/M candidates, then split demand using τ . Reject unsupported candidates before checking latency, throughput, and KV capacity. The target model distinguishes visible first-token latency from P-D continuation behavior. Score feasible plans using operating and standby power plus incremental handoff and switching energy. Re-evaluate the current plan as well; switching to itself costs zero.

用一个反例说明切换门槛 / A switching decision.

图 3 的共同负载含短、长输入各 6 req/s；仅候选 C 按阈值分配到两条路径。A 与 C 均通过合成模型的性能检查；60 秒内 A 消耗 46.148 kJ，C 的运行及停驻能量为 42.497 kJ，另有模型估计的 34 J 切换成本。尚未测得的交接增量非负，因此 C 的归一化总成本至少为 92.16，而 8% 滞回要求低于 92。这一步判断不需要假造未知交接能量，也不意味着其他负载下应拒绝分离。

The workload contains 6 req/s of short prompts and 6 req/s of long prompts. A and C pass the synthetic performance checks. Over 60 seconds, A uses 46.148 kJ; C needs 42.497 kJ for operation and standby plus an estimated 34 J for switching. Unknown extra handoff energy is nonnegative, so C costs at least 92.16 when A is normalized to 100. The 8% margin requires a cost below 92. This pairwise rejection neither

(a) Same forecast: 8 TP1 GPUs; SLO = 1 s / 20 ms; safety factor = 0.85



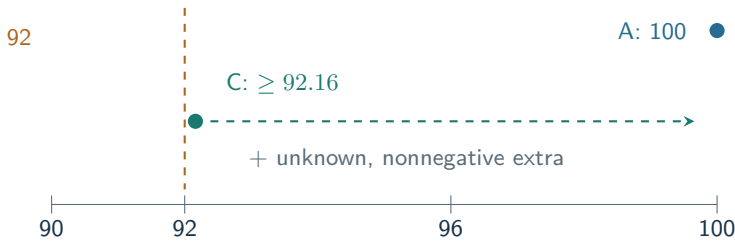
(b) Three displayed candidates; clock labels in MHz

	G0	G1	G2	G3	G4	G5	G6	G7	Power W proxy / TPOT ms
A: current M5 + off3	M 2520	M 2520	M 2520	M 2520	M 2520	off	off	off	769.14 214.19 / 16.03
B: reject M4 + off4	M 2520	M 2520	M 2520	M 2520	off	off	off	off	709.55 218.07 / 17.50
C: feasible P2 + D1 + M2 + off3	P 2100	P 2100	D 2520	M 2520	M 2520	off	off	off	708.28 319.17 / 15.54

Screen: proxy ≤ 850 ms; TPOT ≤ 17 ms. C uses the target search without a fixed M floor.

(c) Same-window GPU energy

$E_A = 46.148$ kJ
 $E_C = 42.497 + 0.034 + E_{\text{extra}}$ kJ
 $E_{\text{extra}} \geq 0$: unknown added handoff energy
 Normalized cost: A = 100; zoom = 90–100



(d) Hysteresis: compare C with A

Margin $\epsilon = 8\%$
 Required gain: 3.692 kJ
 Best-case C gain: ≤ 3.617 kJ

$$3.617 < 3.692 \text{ kJ}$$

$$92.16 > 92$$

C vs A: keep A

Other candidates are not decided.
 Synthetic model; no hardware claim.

Figure 3. 候选筛选与滞回的复算方法例。统一使用仓库合成模型，展示三个候选而非完整搜索。B 的预测 TPOT 超过安全筛选线；C 的运行功率较低，但即使额外交接能量为零，替代 A 的收益仍不足 8%。因此本次 C 对 A 的比较保持 A，不对其他未展示候选作出结论。C 的 M2 配置属于取消固定 M 下界后的目标设计。图中时延列为当前规划模型的 *latency proxy*，并非 carry 协议的实测 TTFT。

A reproducible example of screening and hysteresis. Values come from the repository’s synthetic model. B fails the guarded TPOT constraint. C uses less operating power, but its best-case saving over A is below 8%; this pairwise comparison retains A. Other candidates are not decided here. M2 belongs to the target search without a fixed M floor. Latency proxies are model outputs, not measured carry-protocol TTFTs.

invents missing measurements nor rules out disaggregation under other conditions.

算法 1 的前置条件是 c 已由在线层安全启动，不接受空配置。覆盖检查同时约束候选使用的形状、频率与拓扑；缺测不能解释为零成本。保守回退优先恢复可用容量，必要时限制准入，不保证任意过载下仍满足 SLO。

图中的局部比较用于解释规则，算法则在完整候选集合中选择。

The online layer supplies a ready initial or active plan c ; it cannot be null. Coverage includes shapes, clocks, and topology. Missing measurements do not imply zero cost. Conservative fallback prioritizes available capacity and may limit admission without guaranteeing SLOs under arbitrary over-

Algorithm 1 SELECTIVE-PD-PLAN

Input: Demand F , models M , constraints S , budget B , ready plan c , horizon H , margin ϵ

```
1:  $A \leftarrow \text{CANDIDATES}(B) \cup \{c\}$  ▷ include mixed-only and P/D/M layouts
2:  $V \leftarrow \emptyset$ 
3: for each  $x \in A$  do
4:    $l \leftarrow \text{SPLIT}(F, x)$  ▷ use the candidate's threshold and roles
5:   if not  $\text{COVERED}(M, x, l)$  then continue
6:    $z \leftarrow \text{PREDICT}(M, x, l)$ 
7:   if not  $\text{FEASIBLE}(z, S)$  then continue
8:    $J[x] \leftarrow \text{ENERGY}(z, H) + \text{EXTRAKV}(z, H) + \text{SWITCH}(c, x)$ 
9:    $V \leftarrow V \cup \{x\}$ 
10: end for
11: if  $V = \emptyset$  then return  $\text{SAFE}(c, M, B)$ 
12:  $b \leftarrow \arg \min_{x \in V} J[x]$  ▷ ties prefer mixed-only layouts
13: if  $c \notin V$  then return  $b$ 
14: if  $J[c] - J[b] \leq \epsilon J[c]$  then
15:   return  $c$ 
16: end if
17: return  $b$  ▷ the executor applies the returned Plan
```

load. The figure illustrates a local comparison; the algorithm operates on the full candidate set.

实现对应 / Implementation note. 当前默认候选枚举保留 $M \geq \min(4, \text{slots})$, 目标设计取消这一固定下界, 由覆盖、容量和 SLO 约束选择配置。现有代码已实现枚举、功率预测、切换滞回及阈值输出; 统一验证的增量交接能耗, 以及分别预测可见 TTFT 与续接表现, 仍是目标改进。图中数值来自合成公式, 不构成测量覆盖或硬件性能证据。

The current default search retains at least $\min(4, \text{slots})$ mixed replicas. The target removes this fixed floor and relies on feasibility constraints. Enumeration, power prediction, hysteresis, and threshold output exist today; uniformly validated extra handoff energy and separate visible-TTFT and continuation predictions remain target improvements. Synthetic calculations are not evidence of hardware performance or measured coverage.

4 在线协同与选择性分流

Online Coordination and Selective Routing

4.1 Motivation

离线模型给出服务能力, 规划器给出下一阶段的资源安排, 但请求仍会突发到达。在线层需要持续观察负载, 把计划落实到就绪的引擎, 并在预测偏差扩大前保护延迟。它同时承担观测预测、快速保护、配置执行、逐请求路由和引擎接力, 而不把这些操作留成彼此独立的

策略。

Offline models describe service capacity, and planning proposes the next allocation. Actual requests can still arrive in bursts. The online layer observes demand, applies plans to ready engines, and protects latency before prediction errors accumulate. It coordinates forecasting, protection, resource actions, per-request routing, and engine handoff.

4.2 Challenge

这些操作需要不同的响应速度。新请求必须立即选择去向, 风险需要秒级处理, 而规划、唤醒和排空可能更慢。与此同时, 旧请求仍拥有自己的执行路径和 KV; 提前发布新阈值、重复启动规划或接纳尚未就绪的实例, 都可能使真实负载偏离已评估的方案。

These operations run at different timescales. Routing needs an immediate decision, protection needs a rapid response, and planning, waking, or draining takes longer. Existing requests still own their execution paths and KV. Premature threshold changes, duplicate planning jobs, or unready instances can invalidate the evaluated workload split.

4.3 Insight

使用一份有效配置连接快慢循环: 规划决定资源组合, 保护在覆盖范围内提高能力, 路由只使用已就绪的路径。新阈值与支撑该分流比例的资源一起发布。请求执行和资源动作异步推进, 完成结果与版本回执回到事件循环; 降频和缩容则等待稳定条件, 而不是在压力刚消失时立即发生。

Connect the loops through an effective configuration. Planning selects resources, protection increases covered capacity, and routing uses ready paths. Publish a new threshold together with sufficient resources for its workload split. Requests and resource actions run asynchronously and return events. Reducing capacity requires stable observations, not merely the disappearance of one risk signal.

4.4 Approach

接口与状态。 输入包括请求、性能模型、候选 Plan、在途工作和延迟观测；输出包括 token 流、有效配置、资源动作与完成记录。在线状态区分当前配置、待完成动作和候选计划，同时保留请求绑定的实例与版本。动作完成只说明对应资源已就绪，不能代替请求完成或 KV 释放证明。

Inputs are requests, models, candidate plans, active work, and latency observations. Outputs are token streams, effective configurations, resource actions, and outcomes. Track current configuration, pending actions, and candidate plans separately. An action acknowledgement establishes readiness; it does not establish request completion or KV release.

实例选择与请求接力。 图 4 使用一份独立的已生效配置示例，并非上一章规划图的最终输出。8 个 TP1 实例中，I0 为 P，I1/I2 为 D，I3-I6 为 M，I7 处于 L1。阈值为 1024，两个请求均要求 128 个输出 token。短请求输入 256，进入在途序列最少的 I4；长请求输入 2048，进入唯一 P 实例 I0，并从兼容 D 实例中选择序列更少的 I2。P/D 配对还必须满足模型、拓扑和请求归属的兼容条件。

Figure 4 starts from an independent effective configuration, not the final output of the planning example. Among eight TP1 instances, I0 is P, I1/I2 are D, I3-I6 are M, and I7 is L1. With threshold 1024, the 256-token request chooses the least-loaded M, I4. The 2048-token request chooses I0 and the less-loaded compatible D, I2.

P 执行原始 2048-token 输入并产生真实首 token y_1 。代理先向客户端发出 y_1 ，再以扩展后的 2049-token prompt 和 127-token 剩余预算调用 D。D 复用原始 2048 个位置的 KV，从 y_1 继续生成 $y_2, \dots, y_{128}$ 。最终 usage 仍为输入 2048、输出 128。逻辑 TTFT 截止于客户端收到 P 产生的首 token； y_1 至 y_2 的续接间隙单独观察，不能称为纯 KV 拷贝时间。输出预算为 1 时仅在所选 M 或 P 执行，不建立远程接力，也不占用 D 预留。

P produces the real first token y_1 . The proxy emits it, then calls D with a 2049-token prompt and a remaining budget of 127. D reuses the original 2048 KV positions and continues the same sequence. Combined usage is 2048/128. TTFT ends

when the client receives y_1 ; the continuation gap is separate. One-token outputs use only the selected M or P.

快速保护与稳定回落。 图 5 把抽象闭环展开成一次 decode 压力事件。当前示例周期是 1 秒检查保护、10 秒进行规划，风险变化也能提前请求规划。当 TPOT 的 90 分位达到 SLO 的 0.90 倍并超过 0.80 倍保护门限时，先提高活跃实例频率。风险持续则请求唤醒 I7 作为额外 D；pending 期间 I7 不能接收请求，只有就绪回执通过后才形成 P1+D3+M4 的有效配置。

Figure 5 expands the loop into a decode-pressure episode. Example periods are one second for protection and ten seconds for planning, with earlier replanning on risk changes. A TPOT p90 of 0.90 times the SLO crosses the 0.80 protection threshold. Raise clocks first; sustained risk requests another D. I7 remains closed until its readiness acknowledgement is accepted.

压力消退后，保护期、安静试探和规划稳定条件仍需满足。示例以两个规划窗口确认收缩，再停止 I7 准入、排空并恢复 L1；若压力重现，则恢复所需容量并延长试探窗口。活跃角色复用前也须排空旧工作，或核验新旧负载重叠可行。旧请求不因角色重标记而迁移。图中的频率和阶段仅解释机制，不声称某一时刻已经达到 SLO 或获得能耗收益。

After risk clears, respect protection, quiet probes, and stable planning decisions. The example confirms shrinking across two planning windows, then closes, drains, and parks I7. If risk returns, restore capacity and extend the probe. Reusing an active role also requires draining or validating overlapping work. Existing requests are not migrated by relabeling. Frequencies and stages illustrate behavior, not measured gains.

事件驱动算法。 算法 2 不等待整条请求或资源动作完成；二者通过后台任务返回事件。PLAN 调用上一章的选择性分离规划；重复规划请求合并，只保留最新待评估需求。独立定时器在请求持续到达时仍正常触发。接受结果前检查规划编号与配置版本，接受资源回执前检查动作编号，过期结果不得覆盖新配置。

Algorithm 2 never waits for an entire request or resource action. Background work returns events. PLAN invokes the preceding planning method; duplicate requests are coalesced around the latest demand. Independent timers continue under traffic. Validate planning identifiers and configuration versions, and validate action identifiers before accepting acknowledgements. Stale results cannot replace newer configurations.

PUBLISH-READY 只发布可准入的配置：新阈值须和

Algorithm 2 ONLINE-COORDINATE

```
1: procedure ONLINE-COORDINATE()
2:  $C \leftarrow \text{READY-START}()$ ;  $B \leftarrow C$ ;  $A \leftarrow \emptyset$ 
3: while service is running
4:    $e \leftarrow \text{NEXT-EVENT-OR-TIMER}()$ 
5:    $S \leftarrow \text{OBSERVE}(e)$ 
6:   if request result:  $\text{RECORD}(e)$ 
7:   if action ACK:  $C \leftarrow \text{PUBLISH-READY}(e, A, C)$ 
8:   if new request:  $\text{SPAWN-SERVE}(e.\text{request}, C)$ 
9:    $R \leftarrow \text{UPDATE-PROTECTION}(S)$ 
10:  if slow tick or risk changed
11:     $\text{COALESCE-PLAN}(\text{FORECAST}(S), C)$ 
12:  if plan ready and  $\text{STILL-VALID}(e, S, C)$ 
13:     $B \leftarrow e.\text{plan}$ 
14:   $T \leftarrow \text{STABILITY-FILTER}(B, C, S)$ 
15:   $T \leftarrow \text{PROTECT}(T, R)$ 
16:  if  $\text{NEW-VALID-TARGET}(T, C, A)$ 
17:     $A \leftarrow \text{START-ASYNC-ACTIONS}(C, T)$ 
18:   $\text{HANDLE-FAILED-ACTIONS}(e, A)$ 
```

足够的 M/P/D 路径容量一起生效；过渡期保持旧配置或核验中间配置，必要时限制准入。PROTECT 保持阈值及既有容量下界，只在测量覆盖内升频或扩充角色容量；若改变分流规则则重新评估。稳定条件不能阻挡紧急保护升级。规划与动作异步并不意味着整个集群原子切换。

Publish-Ready installs admissible configurations only: a new threshold requires sufficient M/P/D capacity. Retain the old configuration or validate intermediate states, limiting admission if needed. Protect preserves the threshold and capacity floor while increasing covered clocks or capacity. Routing changes need re-evaluation. Stability gates must not block urgent protection; asynchronous actions do not imply an atomic fleet-wide switch.

Algorithm 3 SERVE: asynchronous request helper

```
1: procedure async SERVE( $r, C$ )
2:  $p \leftarrow \text{NONE}$ ; try
3:    $p \leftarrow \text{VALIDATE-AND-RESERVE}(r, C)$ 
4:   if  $p = \text{NONE}$ : raise unavailable
5:   if  $M$  or  $N = 1$ :  $z \leftarrow \text{await SINGLE}(p, r)$ 
6:   else:  $z \leftarrow \text{await CARRY-AND-DECODE}(p, r)$ 
7: catch failure:  $z \leftarrow \text{FAILED}(\text{failure})$ 
8: finally:  $\text{REPORT-AND-RESOLVE-OWNERSHIP}(r, z, p)$ 
```

算法 3 中 N 为请求的输出预算，派发时再次检查准入和有效版本。短请求的完整输出与长请求的首 token 接力，都在各自异步任务内执行。成功、拒绝、失败和取消仅报告一次；未确认的 KV 或传输资源继续保留归属并隔离，客户端断开不能自动视为资源释放。

In Algorithm 3, N is the output budget. Recheck admission

and version at dispatch. Single-engine output and PD continuation run in their own asynchronous tasks. Report each outcome once. Retain ownership and quarantine uncertain KV or transfer resources; client disconnection is not evidence of release.

实现对应 / Implementation note. 以上版本回执与异步协调是目标设计的组织方式。当前主控制器排空主要依据代理的 prefill 与 sequence 计数，尚不等于已证明原生 KV 及传输资源清零。当前 PD 接力受限于 token-ID 输入、greedy、固定输出预算以及同 TP、PP1；这些限制不应被写成已支持任意采样或跨 TP 迁移。

The acknowledgement-driven organization is the target design. Current main-controller drain primarily uses proxy prefill and sequence counts; this does not prove native KV and transfer resources are empty. Current PD carry requires token IDs, greedy sampling, a fixed output budget, and matching TP with PP1. It does not establish arbitrary sampling or cross-TP migration.

(a) Route two requests under one effective configuration

$8 \times \text{TP1, PP1}$; same model and generation; $\tau = 1024$, $N = 128$

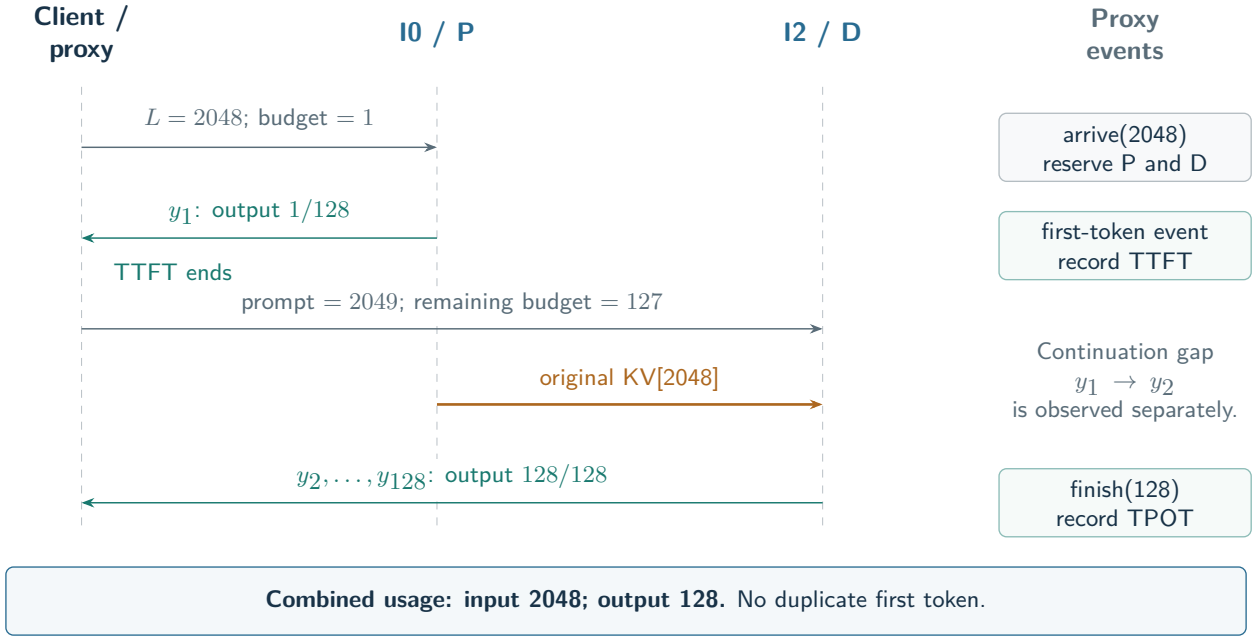
I0 / P pending 4096 tokens	I1 / D 3 active sequences	I2 / D 1 active sequence	I3 / M 2 active sequences	I4 / M 0 active sequences	I5 / M 1 active sequence	I6 / M 3 active sequences	I7 / L1 resident closed
---	--	---------------------------------------	--	--	---------------------------------------	--	--------------------------------------

Rshort: $L = 256$, $N = 128$
 $256 < 1024$; minimum M load: **I4**
I4 executes prefill + all 128 output tokens.

Rlong: $L = 2048$, $N = 128$
 $2048 \geq 1024$; **I0/P + I2/D**
I0 is the only P; D load $1 < 3$ selects **I2**.

Illustrative snapshot; hold other requests unchanged when reading the counters.

(b) Execute Rlong, preserve the first token, and report events

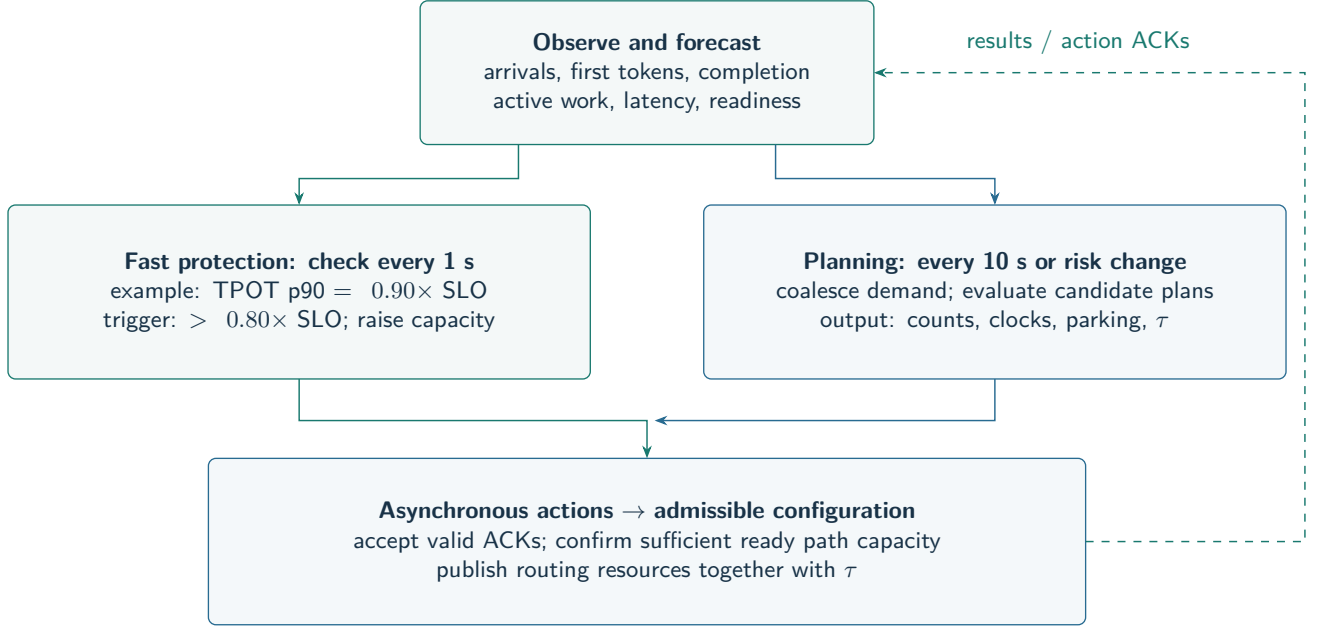


I0 pending tokens: $4096 \rightarrow 6144 \rightarrow 4096$; I2 sequences: $1 \rightarrow 2 \rightarrow 1$.

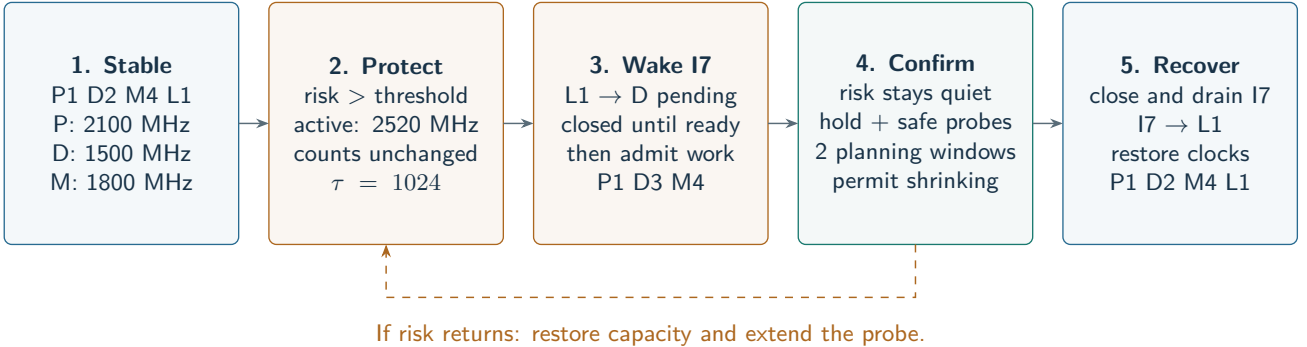
Figure 4. 选择性分流的 worked example. (a) 根据角色、阈值和在途负载选择 I4 或 I0/I2; (b) 保留 P 的首 token, 延长 D 的 prompt 并扣减预算, 再向观测端报告到达、首 token 与完成。示例负载保持其他请求不变; 数字不是实测结果, 箭头表示逻辑依赖而非精确传输重叠。

Worked example of selective routing. (a) Roles, threshold, and active load select I4 or I0/I2. (b) Carry P's first token, extend D's prompt, reduce its budget, and report request events. Other requests are held unchanged in this illustrative snapshot. Values are illustrative; arrows express dependencies rather than exact transfer overlap.

(a) Coordinate independent timers, requests, plans, and action acknowledgements



(b) A conditional state trajectory, not a measured recovery schedule



Requests retain their paths; readiness and drain require evidence. Values are illustrative.

Figure 5. 快慢循环的 worked example. (a) 具体压力信号和候选 Plan 经安全执行进入有效配置；新阈值必须具备足够就绪路径容量。(b) 8 个实例从 P1+D2+M4+L1 经升频和唤醒形成 P1+D3+M4，之后按保护、稳定和排空条件恢复。阶段不对应固定恢复时刻；频率、负载与状态均为示意。

Worked example of online coordination. (a) Pressure and candidate plans reach the effective configuration through safe execution; thresholds require sufficient ready path capacity. (b) Clock escalation and wake-up change P1+D2+M4+L1 to P1+D3+M4, followed by conditional recovery. Stages do not imply fixed recovery times; all values and states are illustrative.